TECHNICAL SOLUTION PITCH

ARIA Full-Duplex Autonomous AI Voice Sales & Support Agent

Production-grade, low-latency conversational telephony engine with domain grounding, anti-starvation audio pacing and real-time dialogue state tracking.

FastAPI | Twilio Media Streams | Gnani AI | Groq LLM | Cloud / Edge | Full-Duplex WebSockets | < 900 ms E2E | 8 kHz μ-law

THE PROBLEM & THE ARIA SOLUTION

INDUSTRY CHALLENGE

- **High telephony latency:** turn-based HTTP webhooks (e.g. Twilio <Gather>) cause awkward 3–5 s pauses and unnatural turn-taking.
- **Audio underflow & choppy voice:** WAN packet jitter starves the phone buffer, causing stutter, clipping and dropped words.
- **ASR misrecognition & phone echo:** loudspeaker echo triggers premature barge-in, while general ASRs mangle Indian accents and brand names.
- **Scripted or hallucinating engines:** decision trees fail on edge queries; unconstrained LLMs invent prices, discounts and policies.

ARIA VALUE PROPOSITION

- **Bidirectional telecom WebSockets:** Twilio Media Streams carry continuous 8000 Hz raw μ-law frames with zero HTTP webhook polling.
- **Paced jitter buffering:** 80 ms chunks, 400 ms lead buffer and 1 ms OS timer resolution give fluid speech with no mid-sentence pauses.
- **Speech intelligence + keyword boosting:** Gnani Prisma STT with custom key-term bias and Gnani Timbre TTS (~280 ms first byte).
- **Grounding guardrails & dynamic personas:** MongoDB-backed agent profiles, strict anti-hallucination limits and 1–2 sentence voice replies.

END-TO-END ARCHITECTURE & CALL PIPELINE

1 Telephony Layer
Twilio Media Stream
WebSocket, 8 kHz μ-law
Full-duplex live stream

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2 Audio & VAD
PCM decoder + RMS VAD
1.0 s silence window
Echo-suppressed mic

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3 Gnani STT
Prisma v3.0 (en-IN)
Vocab boosting: SmartSave, NACH, Dimensity

▶

4 Sales Engine
FastAPI + Groq / Qwen
State-machine tracking
Strict profile grounding

▶

5 Gnani TTS & Out
Timbre v2.5 synthesis
80 ms paced streaming
400 ms jitter lead buffer

CORE TECHNICAL INNOVATIONS

- **Telecom-optimized frame pacing**
  80 ms (640-byte) frames replace 20 ms chunks, cutting network overhead by 75% and preventing ngrok packet clumping.
- **Acoustic echo isolation**
  Suppresses caller loudspeaker feedback while the agent speaks, eliminating premature barge-in cancellations.
- **Tail-end playback sync**
  Waits for physical speaker completion before opening mic VAD, so the agent's last syllable never pollutes transcription.
- **Dynamic multi-agent profiles**
  Instant switching between personalities (SmartSave+ financial advisor, Nova Mobile product consultant) via MongoDB.
- **Live telemetry & dashboard**
  Real-time browser monitor over WebSocket: live 8 kHz waveform, per-component latency and post-call CRM summaries.

PERFORMANCE BENCHMARKS & BUSINESS IMPACT

< 900 ms	100%	0	60–80%
End-to-end turnaround	**Speech continuity**	**Hallucinations**	**Cost reduction**
STT ~400 + LLM ~350 + TTS ~280	400 ms jitter buffer, no underflow	Verified facts, pricing & FAQs only	Autonomous qualification at scale

Rapid integration: plugs into existing CRM systems, Twilio telephony trunking and enterprise product databases. Complex cases are routed to human specialists.

Summary. Aria proves that telecom-grade conversational AI can achieve natural, human fluidity on standard phone lines. By unifying bidirectional WebSockets, native 8 kHz audio streaming, domain-constrained LLMs and acoustic echo cancellation, Aria turns phone calls into high-converting, intelligent customer engagements.